ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C8 · ENERGY

Forms of energy

★☆☆
Grade 4

Learning goals

- Name forms of energy.
- Complete energy chains.

1 Which form of energy? [6]

Example	Form of energy
a flowing river	kinetic
a plate of rice	chemical
a ringing bell	sound
a stretched catapult	stored (potential)
sunshine	light (and heat)
a phone battery	chemical

2 Complete the energy chains. [8]

Device	Energy in	Useful energy out
electric fan	electrical	kinetic
solar panel	light	electrical
radio	electrical	sound
candle	chemical	light and heat

3 Sort the energy sources. [7]

SORT THESE

oil river water wind sunlight natural gas biogas coal

Renewable	Will run out
sunlight	coal
wind	natural gas
biogas	oil
river water	

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C8 · ENERGY

Following the energy

★★☆
Grade 4

Learning goals

- Track energy through systems.
- Explain energy losses.

1 **Draw an energy chain for a rickshaw puller, starting with the Sun.** [3]

Rickshaw energy chain

Sun (light) → rice plant (chemical) → puller's body (chemical) → moving rickshaw (kinetic) + heat.

2 **An old filament bulb gets very hot. An LED bulb stays cool. Which one wastes more energy? Explain.** [2]

The filament bulb: much of its electrical energy becomes heat instead of light.

3 **Rubbing your hands together makes them warm. What energy change is happening?** [1]

kinetic (movement) → heat

4 **Why do we say almost all our energy comes from the Sun? Give two examples.** [2]

Plants store the Sun's energy in food we eat; coal, oil and gas came from ancient plants; wind and rain are driven by the Sun's heat.

5 **Suggest two ways your school could save energy.** [2]

switch off fans and lights at break; use LED bulbs; open windows for daylight; solar panels on the roof.

ANSWER KEY

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INVESTIGATE · C8 · ENERGY

Rubber band launcher

★★☆
Grade 4

Learning goals

- Plan a fair test.
- Plot a line graph.
- Explain using stored and kinetic energy.

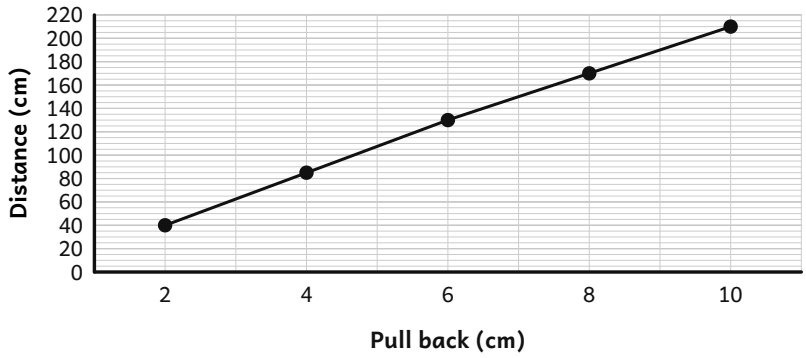
Class 4A used a rubber band stretched between two chair legs to launch a toy car. They pulled the band back by different amounts and measured how far the car travelled.

1 Complete the plan. [3]

Change	how far the band is pulled back (cm)
Measure	distance the car travels (cm)
Keep the same	the same band, car and floor; release without pushing

2 Draw a line graph. [3]

Pull back (cm)	2	4	6	8	10
Distance travelled (cm)	40	85	130	170	210



3 Describe the pattern and explain it using energy. [3]

The further the band is pulled back, the further the car goes, because a more stretched band stores more energy, which becomes more kinetic energy in the car.

4 Predict the distance for a pull of 12 cm. [1]

about 250 cm

ANSWER KEY

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REAL-WORLD CASE STUDY · C8 · ENERGY

Cooking with cow dung: biogas

★★★
Grade 4

Learning goals

- Track energy through a real system.
- Use numbers.
- Weigh up benefits.

Science | Farming | Health | Environment | Economics

In many villages in Bangladesh, families cook on clay stoves that burn wood, leaves or dried cow dung. The smoke fills the kitchen and harms people's lungs, especially women and children.

A **biogas plant** offers another way. Cow dung and water are fed into a closed underground tank. Tiny **microbes** break down the dung and make a gas called **methane**. The gas flows through a pipe to a stove in the kitchen, where it burns with a clean blue flame. The leftover **slurry** is an excellent fertiliser for the fields. A family with a few cows can make enough gas to cook their meals.

Based on information about household biogas in Bangladesh (e.g. IDCOL's biogas programme). Numbers below are examples.

1 **Complete the energy chain for biogas cooking.** [3]

5	gas burns: heat for cooking
1	light from the Sun
4	microbes make biogas
2	grass grows (chemical energy)
3	cows eat grass; dung holds chemical energy

2 **Why is biogas called a renewable energy source?** [2]

As long as cows eat grass and make dung, more gas can be made; it will not run out.

3 **Maths. A family uses 15 kg of firewood a week. A biogas plant replaces all of it. How much wood is saved in a year (52 weeks)?** [2]

15 × 52 = 780 kg of wood

4 **Give one benefit of biogas for each.** [3]

For...	Benefit
health	no smoke in the kitchen, so healthier lungs
the environment	fewer trees cut for firewood
farming	slurry is free fertiliser

ANSWER KEY

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REVIEW AND CHALLENGE · C8 · ENERGY

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

I	S	O	R	E	H	O	Y	W	P
S	W	T	H	O	K	H	E	A	T
B	J	J	O	O	I	F	Y	C	E
I	S	K	N	R	N	I	L	H	P
O	O	E	M	U	E	O	I	A	U
G	U	N	F	S	T	D	G	I	A
A	N	E	Y	O	I	S	H	N	V
S	D	R	J	L	C	J	T	J	D
H	B	G	L	A	N	D	W	C	I
E	G	Y	S	R	P	T	K	L	J

biogas chain energy heat kinetic light solar sound stored

2 What energy change happens in a solar panel? [1]

- ○ heat → sound
- ● **light → electrical**
- ○ electrical → light
- ○ kinetic → chemical

3 Which energy source will run out? [1]

- ○ wind
- ○ sunlight
- ● **natural gas**
- ○ biogas

4 Draw an energy chain for a mobile phone playing music, starting from the charged battery. [3]

Phone energy chain

chemical (battery) → electrical → sound (+ light from the screen, + heat).

5 **Challenge. "Energy is never used up, it only changes form." Explain this using a bicycle that slows down and stops when you stop pedalling.** [2]

The kinetic energy of the bicycle is not destroyed; friction and air resistance change it into heat (and a little sound) in the tyres, brakes and air.